



Figure 1: Peeping through /proc

```

        break;
    case 2:
        len += sprintf(page + len, "time = %ds %dns\n",
                        val / 1000, val % 1000);
        break;
    case 3:
        val /= 1000;
        len += sprintf(page + len, "time = %02d:%02d:%02d\n",
                        val / 3600, (val / 60) % 60, val % 60);
        break;
    default:
        len += sprintf(page + len, "<not implemented>\n");
        break;
}
len += sprintf(page + len, "[offset = %ld; count = %d;]\n",
off, count);

return len;
}

int time_write(struct file *file, const char __user *buffer,
unsigned long count, void *data) {
    if (count > 2)
        return count;
    if ((count == 2) && (buffer[1] != '\n'))
        return count;
    if ((buffer[0] < '0') || ('9' < buffer[0]))
        return count;
    state = buffer[0] - '0';
    return count;
}

static int __init proc_win_init(void) {
    if ((parent = proc_mkdir("anil", NULL)) == NULL) {
        return -1;
    }
    if ((file = create_proc_entry("rel_time", 0666, parent)) ==
NULL) {
        remove_proc_entry("anil", NULL);
        return -1;
    }
    file->read_proc = time_read;
    file->write_proc = time_write;
}

```



Figure 2: Comparison with top's output

```

    if ((link = proc_symlink("rel_time_l", parent, "rel_time")) ==
        NULL) {
        remove_proc_entry("rel_time", parent);
        remove_proc_entry("anil", NULL);
        return -1;
    }
    link->uid = 0;
    link->gid = 100;
    return 0;
}

static void __exit proc_win_exit(void) {
    remove_proc_entry("rel_time_l", parent);
    remove_proc_entry("rel_time", parent);
    remove_proc_entry("anil", NULL);
}

module_init(proc_win_init);
module_exit(proc_win_exit);

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Anil Kumar Pugalia <email@sarika-pugs.com>");
MODULE_DESCRIPTION("Kernel window /proc Demonstration Driver");

```

And then Pugs did the following:

- Built the driver file (*proc_window.ko*) using the usual driver's *Makefile*.
- Loaded the driver using *insmod*.
- Showed various experiments using the newly created proc windows. (Refer to Figure 1.)
- And finally, unloaded the driver using *rmmmod*.

Demystifying the details

Starting from the constructor *proc_win_init()*, three proc entries have been created:

- Directory *anil* under */proc* (i.e., NULL parent) with default permissions 0755, using *proc_mkdir()*
- Regular file *rel_time* in the above directory, with permissions 0666, using *create_proc_entry()*
- Soft link *rel_time_1* to the file *rel_time*, in the same directory, using *proc_symlink()*

The corresponding removal of these is done with